

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

NAVIGATION CONTROL MODULE (NCM) - INDIA & ISRAEL

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable.
Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Navigation Control Module (NCM) - India & Israel. For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

For additional information, refer to: [Navigation System](#) (415-01 Information

and Entertainment System, Description and Operation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B119F-31	GPS Antenna - No Signal	GPS antenna circuit short circuit to ground, short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the GPS antenna circuit for short circuit to ground, short circuit to power, open circuit, high resistance
B1296-4A	Navigation Map Data Mismatch With Market - Incorrect component installed	SD memory card map data incorrect to market	Install the correct SD memory card
B1A81-4B	Internal Trip Switch - Over temperature	Navigation control module internal temperature above threshold	Allow the vehicle to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new navigation control module
B1A89-11	Satellite Antenna - Circuit short to ground	GPS antenna circuit short circuit to ground	Refer to the electrical circuit diagrams and check the GPS antenna circuit for short circuit to ground
B1A89-13	Satellite Antenna - Circuit open	GPS antenna circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the GPS antenna circuit for open circuit, high resistance
U0010-88	Medium Speed CAN Communication Bus - Bus off	Medium speed CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance

		resistance	
U0011-82	Medium Speed CAN Communication Bus Performance - Alive/sequence counter incorrect/not updated	Invalid data received from another control module via the medium speed CAN bus (comfort)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0011-87	Medium Speed CAN Communication Bus Performance - Missing message	Missing message from another control module via the medium speed CAN bus (comfort)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index